

Signal Processing: Methods and Algorithms

Laboratory 7

1 Simulating the motion of a boat

We model the motion of a boat subject to a constant wind force by the discrete-time random system

$$\underline{x}[n] = \Phi \underline{x}[n-1] + \underline{b}[n-1] + \underline{\eta}[n-1] \quad (1)$$

where $\underline{x}[n]$ is a 4×1 vector representing the position and velocity of the boat on the ocean, Φ is the 4×4 transition matrix, $\underline{b}[n-1]$ is a 4×1 vector that models the constant wind force (corresponding to the input $\underline{bu}[n-1]$ used in the lecture), and $\underline{\eta}[n-1]$ is a 4×1 vector of Gaussian random variables with zero mean and covariance matrix Q which models the random forces due to the waves acting on the boat.

Simulate $N = 100$ samples of the state vector $\underline{x}[n]$ by using Eq. (1) with

$$\Phi = \begin{bmatrix} 1 & 0.6321 & 0 & 0 \\ 0 & 0.3679 & 0 & 0 \\ 0 & 0 & 1 & 0.6321 \\ 0 & 0 & 0 & 0.3679 \end{bmatrix}, \quad \underline{b}[n-1] = \begin{bmatrix} 0.3679 \\ 0.6321 \\ 0.3679 \\ 0.6321 \end{bmatrix}, \quad Q = \begin{bmatrix} 0.0168 & 0.0200 & 0 & 0 \\ 0.0200 & 0.0432 & 0 & 0 \\ 0 & 0 & 0.0168 & 0.0200 \\ 0 & 0 & 0.0200 & 0.0432 \end{bmatrix} \quad (2)$$

and by taking zero initial conditions

$$\underline{x}[0] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (3)$$

Note: To simulate one sample of the vector $\underline{\eta}[n-1]$ of Gaussian random variables with zero mean and covariance Q , use the following Matlab code

```
mu=zeros(4,1);
eta=mvnrnd(mu,Q).';
```

Generation of the measurements. The position of the boat on the x and y axis is measured by a GPS receiver. The measurements are modeled by

$$\underline{z}[n] = H \underline{x}[n] + \underline{v}[n] \quad (4)$$

where $\underline{z}[n]$ is the 2×1 vector of measurements, H is the measurement matrix given by

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (5)$$

and $\underline{v}[n]$ is the measurement noise, modeled by a 2×1 vector of Gaussian random variables with zero mean and covariance matrix

$$R = r \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (6)$$

Simulate the vector of measurements $\underline{z}[n]$ by taking $r = 10$. Note: To simulate N samples of the measurement noise $\underline{v}[n]$ use the following Matlab code

```
v=sqrt(r)*randn(2,N);
```

Plot the following signals in the same Matlab figure:

1. The boat position on the x axis (first component $x_1[n]$ of the state vector $\underline{x}[n]$) with respect to the boat position on the y axis (third component $x_3[n]$ of the state vector $\underline{x}[n]$).
2. The measurements of the boat position on the x axis (first component $z_1[n]$ of the measurement vector $\underline{z}[n]$) with respect to the measurements of the boat position on the y axis (second component $z_2[n]$ of the state vector $\underline{z}[n]$).

2 Kalman filter

Generate the estimate $\hat{\underline{x}}[n]$ of the boat position and velocity $\underline{x}[n]$ by using a Kalman filter whose input are the measurements $\underline{z}[n]$. The function `Kalman_BoatWithWind`, which implements the Kalman filter, is given. The syntax of the function is the following:

```
[xenp,Pnp] = Kalman_BoatWithWind(zn,xen_1p,Pn_1p,H,R,PHI,Q,bn_1)
```

where:

- **xenp** is the updated estimate $\hat{\underline{x}}^+[n]$ obtained by combining $\hat{\underline{x}}^-[n]$ and the measurement $\underline{z}[n]$ ($\hat{\underline{x}}^+[n]$ is the required estimate $\hat{\underline{x}}[n]$).
- **Pnp** is the updated error covariance matrix $P^+[n]$ (the error covariance matrix corresponding to the estimate $\hat{\underline{x}}[n]$).
- **zn** is the measurement $\underline{z}[n]$ at time n .
- **xen_1p** is the previous state estimate $\hat{\underline{x}}^+[n-1]$ (that is $\hat{\underline{x}}[n-1]$).
- **Pn_1p** is the error covariance matrix $P^+[n-1]$ (the error covariance matrix of the previous estimate $\hat{\underline{x}}[n-1]$).
- **H** is the measurement matrix H .
- **R** is the covariance matrix R .
- **PHI** is the transition matrix Φ .
- **Q** is the covariance matrix Q .
- **bn_1** is the vector $\underline{b}[n-1]$.

The Kalman filter must be used in an iterative form, that is, it must use one sample of the measurements $\underline{z}[n]$ at a time. Set up a **for** cycle that feeds $\hat{\underline{x}}[n-1]$ and $P^+[n-1]$ to the Kalman filter, and obtains $\hat{\underline{x}}[n]$ and $P^+[n]$. Choose zero initial conditions for the state estimate $\hat{\underline{x}}[0]$ and a zero error covariance matrix $P^+[0]$.

Plot the following signals in the same Matlab figure:

1. The boat position on the x axis (first component $x_1[n]$ of the state vector $\underline{x}[n]$) with respect to the boat position on the y axis (third component $x_3[n]$ of the state vector $\underline{x}[n]$).
2. The measurements of the boat position on the x axis (first component $z_1[n]$ of the measurement vector $\underline{z}[n]$) with respect to the measurements of the boat position on the y axis (second component $z_2[n]$ of the state vector $\underline{z}[n]$).
3. The estimated boat position on the x axis (first component $\hat{x}_1[n]$ of the state estimate $\hat{\underline{x}}[n]$) with respect to the estimated boat position on the y axis (third component $\hat{x}_3[n]$ of the state vector $\hat{\underline{x}}[n]$).